

Line Follower Robot - Full Wiring Connections

1. Power Connections

Component	Connection
Arduino Nano	USB or Battery (5V-7.4V)
TB6612FNG Motor Driver	VCC -> 5V (Arduino Nano)
	VM -> Battery Positive (6V-9V)
	GND -> Battery Negative & Arduino GND
IR Sensor Array (7 Sensors)	VCC -> 5V (Arduino Nano), GND -> GND

2. TB6612FNG Motor Driver to Arduino Nano

TB6612FNG	Arduino Nano
STBY	5V
PWMA	D5 (PWM)
AIN1	D4
AIN2	D3
PWMB	D6 (PWM)
BIN1	D7
BIN2	D8

3. N20 Motors to TB6612FNG

TB6612FNG	Motor
A01, A02	Left Motor
B01, B02	Right Motor

4. 7-Channel IR Sensor Array to Arduino Nano

IR Sensor	Arduino Nano
Sensor 1	A0
Sensor 2	A1
Sensor 3	A2
Sensor 4	A3
Sensor 5	A4
Sensor 6	A5
Sensor 7	A6

5. Additional Components

LED Indicator

LED	Arduino Nano
LED	D13
GND	GND

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Push Buttons (For Modes)

Button	Arduino Nano
Calibration Button	D11
Start Button	D12
GND (Both Buttons)	GND

6. Caster Wheel & Chassis Assembly

- Mount the caster wheel at the back for balance.
- Secure the motors and sensor array properly.

Notes

- D11 -> Calibration Mode (Reads sensor values, adjusts thresholds, tests motor movement)
- D12 -> Start Mode (Begins the maze-solving logic)
- D13 -> LED Indicator for status